SIMATS ENGINEERING ASSESSMENT TOOL

Design Task

COURSE NAME: Theory of Computation COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO2: De**ign Finite Automata DFA, NFA, ε-NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)**

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:25 Duration : 1 Hour

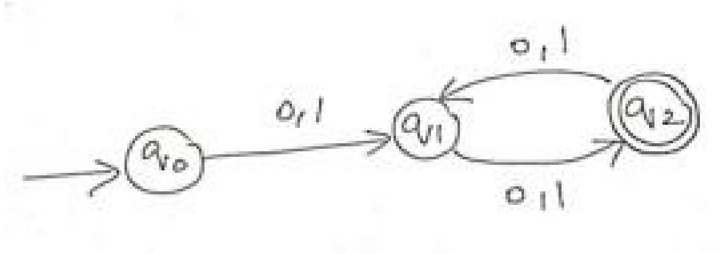
Code of Conduct:

I ***Yuvann Nithish*** **R (192411267)** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary

action.

Signature of the student: Yuvann Nithish R

1.Design the regular expression for the language represented by the given



automata

1. Consider the following grammar

S->aSc|T|U|e

T-> bTc|e

U->aUb|e

A} What is the language defined by the grammar?

B)Give the derivation tree for the word aaabbc

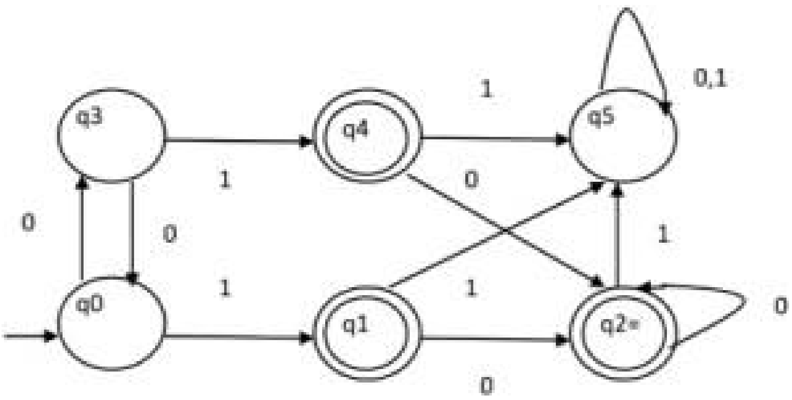
C)A Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify.

2. Design a non deterministic finite automata with ε-transitions to represent the language denoted by the regular expression
00*1+11*0

3. **I)**Show that if L1 and L2 are regular then (L1+l2) also regular

Ii)Is the language { a^p| p is prime} regular? Justify

4. Design minimized automata, Consider the DFA given below. Find the states that are equivalent and construct a minimum state automata equivalent to the given automata. Draw the minimized automata, write its transition table and mark initial and final states



1. Design the Regular Expression for the given automaton

Automaton

- Start State: **q0**
- Final State: **q2**
- q0 → q1 on **0,1**
- q1 ↔ q2 on **0,1**

The automaton accepts all binary strings of **even length (≥2)**.

Regular Expression

$$\boxed{((0+1)(0+1))^{+}}$$

or equivalently,

$$\boxed{(0+1)(0+1)((0+1)(0+1))^{*}}$$

2. Consider the Grammar

$$S \to aSc \mid T \mid U \mid \varepsilon$$
$$T \to bTc \mid \varepsilon$$
$$U \to aUb \mid \varepsilon$$

(A) Language Defined

The grammar generates

- $a^n c^n$
- $b^n c^n$
- $a^n b^n$

including the empty string.

Hence,

$$\boxed{L = \{a^n c^n\} \cup \{b^n c^n\} \cup \{a^n b^n\},\ n \geq 0}$$

(B) Derivation for aaabbcc

Leftmost derivation:

S
⇒ aSc
⇒ aaScc
⇒ aaaUcc
⇒ aaaUbcc
⇒ aaabbcc

Parse Tree

```
      S
     / | \
    a  S  c
      /|\
     a S c
      /|\
      a U c
```

```
 /|\
b U
  |
  b
  |
  ε
```

(C) Is the grammar ambiguous?
Yes.
Reason:
The string **ε** has multiple derivations.
S⇒ε
and
S⇒T⇒ε
and
S⇒U⇒ε
Since one string has more than one derivation tree, the grammar is **ambiguous**.

3. Design an ε-NFA for

$$00^{*}1 + 11^{*}0$$

This represents

$$0^{+}1 \ \cup \ 1^{+}0$$

ε-NFA

```
     ε
 → q0------ → q1 --0--> q1 --1--> q2(F)
  |
  |
  ------→ q3 --1--> q3 --0--> q4(F)
```

Transition Table

State	**ε**	**0**	**1**
q0	q1,q3	-	-
q1	-	q1	q2
q2	-	-	-
q3-----------------			q3
q4	-	-	-

Final states = **q2, q4**

4(i). Show that if L_1 and L_2 are regular then L_1+L_2 is regular
Since L_1 and L_2 are regular, there exist DFAs M_1 and M_2.
Construct an NFA by introducing a new start state with ε-transitions to M_1 and M_2.
The NFA accepts

$$L_1 \cup L_2$$

Since every NFA has an equivalent DFA,

$$\boxed{L_1 + L_2 \text{ is regular.}}$$

4(ii). Is

$$L = \{a^p \mid p \text{ is prime}\}$$

regular?

Answer: No

Using the Pumping Lemma,

Assume L is regular.

Take

$$w = a^p$$

where $p$is a prime greater than the pumping length.

After pumping,

$$xy^2z$$

produces a string whose length is no longer prime.

This contradicts the Pumping Lemma.

Hence,

$$\boxed{L = \{a^p \mid p \text{ is prime}\} \text{ is NOT regular.}}$$

5. DFA Minimization

Initial States

Start : q0

Final States :

q1

q2

q4

Equivalent States

Using partition refinement,

- **q1 ≡ q4**
- **q2 ≡ q5**

Therefore,

Merge

A = {q1,q4}

B = {q2,q5}

Remaining states

q0

q3

A={q1,q4}

B={q2,q5}

Minimized DFA

State	0	1	Final
q0	q3	A	No
q3	q0	A	No
A	B	B	Yes
B	B	B	Yes

Initial State

→ q0

Final States

* A={q1,q4}

* B={q2,q5}

Minimized DFA Diagram (Text)

```
          1
   →q0-------- > A*
    |          |
   0|          |0,1
    v          v
   q3 ----1----> B*
    |          ^
    |0         |0,1
+     --+---------
```

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand